

Lysosomes in senescence and aging

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Abstract

Dysfunction of lysosomes, the primary hydrolytic organelles in animal cells, is frequently associated with aging and age-related diseases. At the cellular level, lysosomal dysfunction is strongly linked to cellular senescence or the induction of cell death pathways. However, the precise mechanisms by which lysosomal dysfunction participates in these various cellular or organismal phenotypes have remained elusive. The ability of lysosomes to degrade diverse macromolecules including damaged proteins and organelles puts lysosomes at the center of multiple cellular stress responses. Lysosomal activity is tightly regulated by many coordinated cellular processes including pathways that function inside and outside of the organelle. Here, we collectively classify these coordinated pathways as the lysosomal processing and adaptation system (LYPAS). We review evidence that the LYPAS is upregulated by diverse cellular stresses, its adaptability regulates senescence and cell death decisions, and it can form the basis for therapeutic manipulation for a wide range of age-related diseases and potentially for aging itself.

Keywords age-related disease; autophagy; LYPAS; lysosomal quality control; senescence

Subject Categories Autophagy & Cell Death; Molecular Biology of Disease; Organelles

DOI 10.15252/embr.202357265 | Received 29 March 2023 | Revised 8 August 2023 | Accepted 21 September 2023 | Published online 9 October 2023

EMBO Reports (2023) 24: e57265

Introduction

For many years, the lysosome had a self-esteem problem. While mitochondria and other organelles were given cool nicknames such as the “powerhouse of the cell”, the lowly lysosome languished. This emotional distress likely could be traced back to their initial discovery, when Christian De Duve gave them the unfortunate moniker, “suicide bags” of the cell (De Duve *et al.*, 1955; de Duve, 1959). Lately, however, a confluence of events has led to lysosomes getting a much-needed makeover. Through their broad degradative activity from dozens of hydrolytic enzymes, lysosomes are increasingly viewed as being uniquely positioned to be at the center of diverse processes including cellular metabolism, nutrient sensing, ion

homeostasis, and immunity. Lysosomal degradation is now viewed beyond its narrowly constrained role in housekeeping and nutrient recycling. Increasingly, lysosomes are viewed as central to critical cellular quality control organelles that provide adaptation to a wide range of cellular stresses, including infection, protein aggregation, organelle damage, and oxidative stress.

Lysosomal dysfunction is a common factor associated with many human diseases. Inherited mutations severely compromising lysosomal activity underlie an array of congenital lysosomal storage diseases (LSD) often resulting in early mortality (Platt *et al.*, 2018), whereas mutations that more modestly affect lysosomal activity appear to contribute to late-onset diseases characterized by impaired endo-lysosomal function (Carmona-Gutierrez *et al.*, 2016). Lysosomal dysfunction can trigger profound cellular changes such as increased oxidative stress, accumulation of damaged macromolecules and organelles, and augmented lysosomal secretion.

Cellular senescence is an evolutionarily conserved process with important physiological roles in tissue remodeling during development and in wound healing (Muñoz-Espín & Serrano, 2014). Senescence is also one of the hallmarks of aging and age-related diseases, a unique cellular state characterized by permanent cell cycle arrest, increased macromolecule damage, and the senescence-associated secretory phenotype (SASP), among other features (Chinta *et al.*, 2015; Baker & Petersen, 2018; Sikora *et al.*, 2021). Profound lysosomal changes are found in senescent cells, including dramatic expansion in lysosomal size and number, pH neutralization, lysosomal membrane permeabilization (LMP), accumulation of undigested auto-fluorescent material known as lipofuscin, and upregulation of lysosomal enzymes such as β -galactosidase (Lee *et al.*, 2006; Höhn & Grune, 2013; Gómez-Sintes *et al.*, 2016; Pu *et al.*, 2016; Platt *et al.*, 2018). Indeed, the histological stain, senescence-associated β -galactosidase (SA- β gal), perhaps the most widely employed marker of the senescent state, largely reflects an increase in lysosomal content (Kurz *et al.*, 2000; Lee *et al.*, 2006; Debacq-Chainiaux *et al.*, 2009). In neurodegenerative diseases where large-scale neuronal cell death is commonly observed, affected neurons also exhibit an expansion of dysfunctional lysosomes (Piras *et al.*, 2016). However, it has been unclear whether lysosomal alterations represent a driver or consequence of senescence (Gorgoulis *et al.*, 2019). Although directly compromising lysosomal activity or integrity can induce senescence or cell death, other cellular insults seemingly unrelated to lysosomes also

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trigger senescence or cell death accompanied by subsequent lysosomal expansion and dysfunction.

In this review, we introduce the concept of the lysosomal processing and adaptation system (LYPAS) to incorporate both the lysosome itself and multiple interconnected cellular processes that help maintain lysosomal substrate processing capacity for cellular homeostasis and stress resolution. We discuss how this system responds to diverse cellular perturbations and how overloading and exhaustion of this system can impact senescence or cell death. Finally, we discuss how the prevention of LYPAS overload or exhaustion can act as a disease treatment strategy and a means of improving overall healthspan.

The lysosomal processing and adaptation system in health and disease

Optimal lysosomal function is achieved by many interconnected, highly organized cellular processes constituting the LYPAS. Defects in any LYPAS components can cause an array of pathological conditions including LSD, premature aging, cardiovascular diseases, and neurodegeneration. Therefore, when assessing the role of lysosomes in health and disease, it is useful to consider the LYPAS system as a whole. We propose that the LYPAS consists of multiple components falling into five major categories (Fig 1): (i) basal lysosomal maintenance; (ii) lysosomal substrate delivery; (iii) lysosomal small molecule transport and osmoregulation; (iv) lysosomal membrane dynamics; and (v) lysosomal quality control.

Basal lysosomal maintenance

The basal maintenance of lysosomal activity is supported by correct lysosomal targeting of active hydrolytic enzymes, co-enzymes, and resident lysosomal membrane proteins and by V-ATPase-mediated lysosomal acidification (Fig 1). Lysosomal membrane proteins typically carry lysosomal sorting signals/motifs that are recognized at the trans-Golgi network (TGN) by vesicle trafficking adaptor factors that target the subsequent protein trafficking to the endolysosomal compartment (Bonifacino & Traub, 2003). Most soluble lysosomal enzymes undergo mannose 6-phosphate (M6P) modification at the Golgi complex and are then captured by the M6P receptors at the TGN for their lysosomal targeting (Braulke & Bonifacino, 2009). Correct synthesis and localization of lysosomal proteins are fundamental for lysosomal activity. Mutations in genes encoding lysosomal enzymes or membrane proteins or their trafficking factors are the underlying causes of many LSDs (Platt *et al*, 2018). For instance, mutations affecting key factors of the M6P lysosomal enzyme trafficking pathway are known to cause a group of lysosomal storage disorders known as the mucopolysaccharidoses (Khan & Tomatsu, 2020). Similarly, the lysosomal enzyme trafficking factor (LYSET, also known as GCAF or TMEM251), binds to, stabilizes and/or activates N-acetylglucosaminyl-1-phosphotransferase (GNPT), the first key enzyme in M6P tagging (Pechincha *et al*, 2022; Richards *et al*, 2022; Zhang *et al*, 2022). Pathogenic mutations in LYSET cause a severe type of LSD resembling GNPT deficiency (Richards *et al*, 2022).

As most lysosomal hydrolytic enzymes require an acidic pH for optimal activity, acidification is an essential component of lysosomal integrity. Lysosomal acidification is mediated by the proton pump V-type ATPase (V-ATPase), a protein complex encoded by

more than a dozen ATP6 genes that transport protons into the lysosomal lumen using the energy from ATP hydrolysis (Ohkuma *et al*, 1982). The V-ATPase consists of a soluble ATP-hydrolyzing V1 subcomplex and a membrane-embedded V0 subcomplex (Mindell, 2012; Abbas *et al*, 2020; Vasanthakumar & Rubinstein, 2020; Wang *et al*, 2020a, 2020b). The disassembly and reassembly of the V1/V0 subcomplexes are dynamically regulated by lysosomal pH status and degradation needs (Kane, 1995; Mindell, 2012). Lysosomal pH is further balanced by the proton-activated proton channel TMEM175 which releases proton from the lysosome at low pH to avoid over acidification (Hu *et al*, 2022a; Zheng *et al*, 2022b). Lysosomal-associated membrane protein 1 (LAMP1) and LAMP2, widely used as lysosomal markers, directly bind to TMEM175 and inhibit its channel activity, allowing for efficient lysosomal acidification (Zhang *et al*, 2023). Healthy aging and extended lifespan correlate with more efficient lysosomal acidification (Hughes & Gottschling, 2012; Molin & Demir, 2014; Ruckenstein *et al*, 2014; Sun *et al*, 2020). Inherited mutations in V-ATPase are known to cause or contribute to a number of human diseases, such as Cutis laxa (CL), wrinkly skin syndrome (WSS), distal renal tubular acidosis, Parkinson's and Alzheimer's diseases, as well as other neurodegenerative disorders (Jin *et al*, 2012; Lee *et al*, 2022). Lysosomal acidification also declines in normal aging and age-related diseases (Colacurcio & Nixon, 2016).

Lysosomal substrate delivery

Efficient substrate capture and delivery are the prerequisites of lysosomal degradation. Based on capturing approaches and delivery mechanisms, at least four distinct lysosomal delivery routes are available (Fig 1): endocytosis, macroautophagy, microautophagy, and chaperon-mediated autophagy (CMA) (Yim & Mizushima, 2020). Here, endocytosis refers to all forms of internalization including but not limited to receptor-mediated endocytosis, pinocytosis, and phagocytosis (Conner & Schmid, 2003). Through complementary substrate delivery mechanisms, cells degrade various macromolecules in the lysosome for nutrient recycling and cellular homeostasis.

While the total endocytosis rates are reduced in aging, caveolin-mediated endocytosis appears to be upregulated in senescence which in neurons appears to drive prion-like propagation of aberrant α -synuclein oligomers (Ha *et al*, 2021), a key step in the progression of synucleinopathies such as Parkinson's disease (Krüger *et al*, 1998; Stefanis, 2012). Microtubule-associated protein 1A/1B light chain 3 (LC3)-associated phagocytosis (LAP) and endocytosis (LANDO) are emerging as critical routes for lysosomal substrate delivery which are impaired in aging and age-related disease (Sanjuan *et al*, 2009; Martinez *et al*, 2015; Heckmann *et al*, 2019, 2020; Inomata *et al*, 2020). Most other lysosomal delivery pathways also exhibit diminished activity in organismal aging, especially macroautophagy, microautophagy, and CMA (Cuervo & Dice, 2000a, 2000b; Kiffin *et al*, 2007; Sun *et al*, 2015; Barbosa *et al*, 2018; Krause *et al*, 2022). As a master regulator of macroautophagy, mammalian target of rapamycin complex 1 (mTORC1) activated on lysosomes directly phosphorylates and suppresses autophagy-initiating proteins, most notably ULK1, ULK2, and ATG13 (Chang & Neufeld, 2009; Ganley *et al*, 2009; Hosokawa *et al*, 2009; Jung *et al*, 2009; Kim *et al*, 2011). The inactivation of mTORC1-dependent nutrient signaling is a key mechanism for the upregulation of

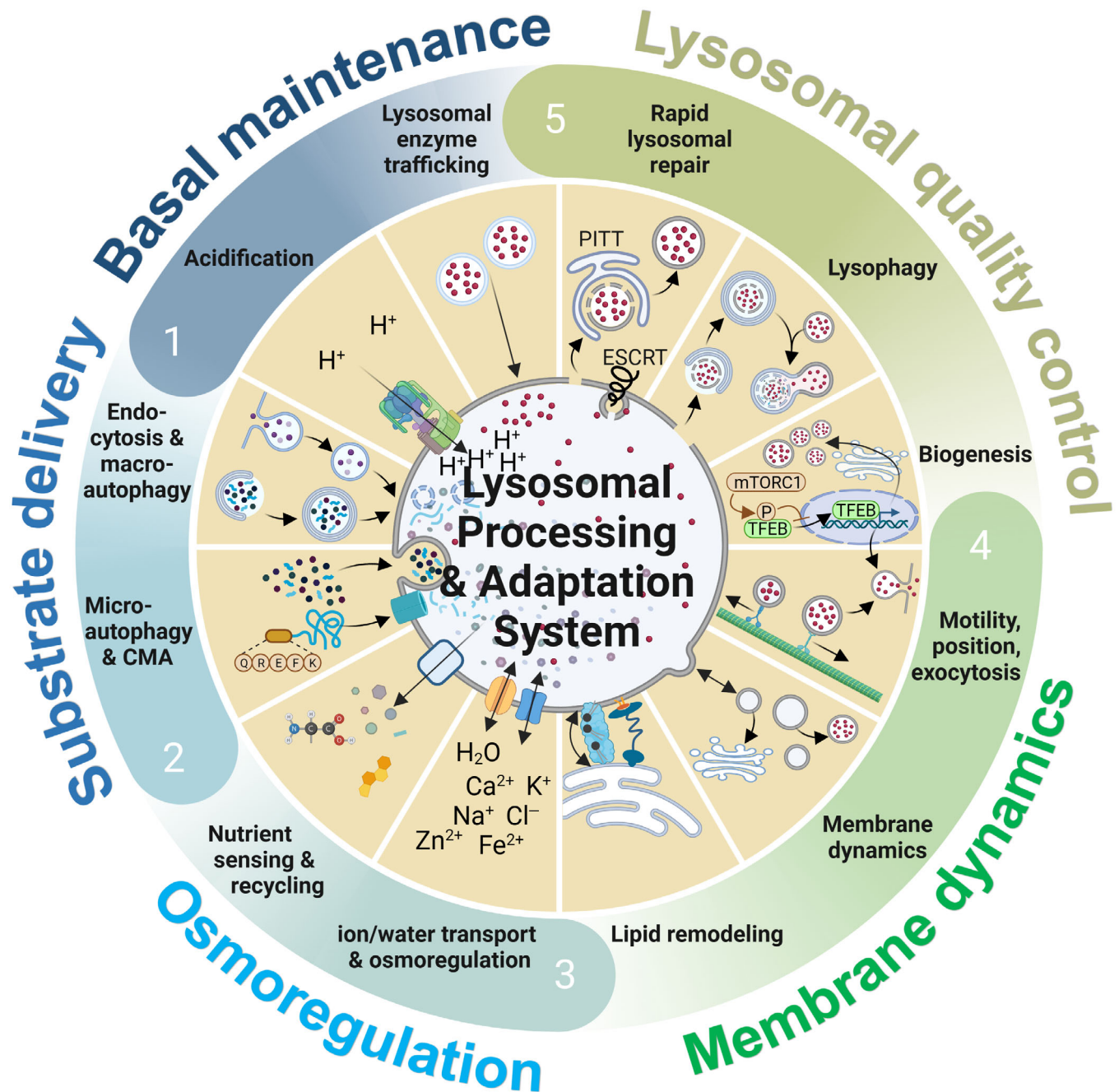


Figure 1. The lysosomal processing and adaptation system (LYPAS).

Lysosomal function is maintained by the entire LYPAS depicted here as 12 integrated components within five major categories. All LYPAS components are critical for the proper function of lysosomes. Upregulating lysosomal activity involves the coordinated mobilization of various LYPAS components. Graphics were created with BioRender.com.

macroautophagy, as extensively reviewed recently (Kim & Guan, 2019; Wang & Zhang, 2019; Liu & Sabatini, 2020). Conversely, activation of mTORC1 appears to be a major factor contributing to the decreased macroautophagic flux observed with aging (Herranz *et al*, 2015; Johnson *et al*, 2015; Liu & Sabatini, 2020). In addition, transcriptional downregulation of various macroautophagy components is also observed in aging (Lipinski *et al*, 2010; Omata

et al, 2014; Ott *et al*, 2016a; Aman *et al*, 2021), which likely involves age-dependent loss of autophagy-promoting transcription factors, as well as epigenetic modulation of macroautophagy genes (Zheng *et al*, 2018; Zhang *et al*, 2019; Wong *et al*, 2020; Wang *et al*, 2021). The expression of LAMP2A, which is essential for substrate translocation into lysosomes during CMA, is also reduced with age (Cuervo & Dice, 2000a). Genetic or pharmacological restoration of CMA

appears to forestall a host of age-related pathologies (Zhang & Cuervo, 2008; Kaushik & Cuervo, 2018; Bourdenx *et al.*, 2021). In general, malfunction of lysosomal substrate delivery causes cellular accumulation of damaged macromolecules and organelles that can in turn trigger senescence or cell death. As such, blocking macroautophagy or CMA by deleting essential autophagy proteins or LAMP2, respectively, induces cellular senescence (Kang *et al.*, 2011).

Lysosomal small molecule transport and osmoregulation

Lysosomal small molecule transport and osmoregulation are integral parts of lysosomal physiology. Lysosomes have extensive cross-membrane transport of small molecules partially due to their major role in hydrolytic breakdown that generates tremendous amounts of metabolites including amino acids, nucleotides, and lipids, all of which need to be transported out of the lysosome for both cell signaling and nutrient recycling purposes (Fig 1). During lysosomal hydrolytic degradation, water is consumed with the generation of organic hydrolytic products, which dramatically increases lysosomal osmolarity (i.e. the total number of solutes in a given volume of the lysosomal lumen) causing water influx, lysosomal swelling, and increased membrane tension (Hu *et al.*, 2022b). The timely reduction of lysosomal osmotic stress/membrane tension is required for efficient lysosomal membrane remodeling such as budding, fission, and intraluminal sorting (Saric & Freeman, 2021). The lysosomal lumen contains various ions, including H^+ , Na^+ , K^+ , Ca^{2+} , Cl^- , Fe^{2+} , and Zn^{2+} . Lysosomal membrane channels and transporters for organic molecules, ions, and water coordinate in rebalancing lysosomal osmolarity and adjusting organelle membrane tension (Scott & Gruenberg, 2011; Saric & Freeman, 2021). Thus, activation of these small molecule transporters is expected in response to cellular stresses, such as nutrient starvation, oxidative stress, or protein damage, when lysosomal activity is correspondingly increased. Failure to efficiently adjust lysosomal osmolarity may result in lysosomal swelling, membrane rupture, or in the extreme case, LSDs. Examples of transporters whose mutations cause LSDs include the lysine/arginine transporter SLC66A1/PQLC2 (Liu *et al.*, 2012), the cystine transporter SLC66A4/cystinosin (Kalatzis *et al.*, 2001), and the sialic acid transporter SLC17A5/sialin (Verheijen *et al.*, 1999). Mutation of the lysosomal Ca^{2+} channel TRPML1/mucopolipin-1 results in **Mucopolipidosis type IV** (MLIV), a rare LSD characterized by vision loss and intellectual disability (Khan & Tomatsu, 2020).

Lysosomal membrane dynamics

Lysosomal activity is further tuned by lysosomal membrane dynamics including lipid signaling, lipid metabolism, lipid transfer, membrane contacts, membrane remodeling, as well as lysosomal motility, positioning, and exocytosis (Fig 1). Phosphatidylinositol 3,5-bisphosphate [PI(3,5)P₂], a lipid messenger produced by the FYVE finger-containing phosphoinositide kinase (PIKfyve), plays an essential role in lysosomal membrane homeostasis through the regulation of lysosomal membrane tubulation, ion channel activation or inhibition, and lysosomal acidification (Li *et al.*, 2014; Chen *et al.*, 2017; Hirschi *et al.*, 2017; Schmiede *et al.*, 2017; She *et al.*, 2018; Rivero-Ríos & Weisman, 2022). Genetic mutations of PIKfyve or its interacting partners cause many human diseases, particularly neurodegenerative diseases (Huang *et al.*, 2021). Another lipid messenger PI(4)P is also produced on lysosomes but is dynamically removed

by oxysterol-binding protein (OSBP)-related proteins (ORP) for hydrolysis on the endoplasmic reticulum (ER) in exchange for ER-to-lysosome transfer of cholesterol or phosphatidylserine (PS) (Dong *et al.*, 2016; Lim *et al.*, 2019; Kawasaki *et al.*, 2021; Tan & Finkel, 2022). Such ER-lysosome lipid exchanges are essential in the regulation of lysosomal lipid composition and rapid lysosomal repair. Additional lysosomal signaling phosphoinositides that include PI(3)P and PI(4,5)P₂ have been recently reviewed (Ebner *et al.*, 2019).

Lysosomes are active centers for lipid metabolism. Lysosomal lipid processing and export are critical for lysosomal integrity and activity. A well-known example is lysosomal cholesterol egress, the disruption of which causes Niemann-Pick disease type C (NPC). Cholesterol extracted from endocytosed lipoprotein complexes is transported by NPC2 and NPC1 from the lysosomal lumen to the limiting membrane (Qian *et al.*, 2020). After this, cholesterol is further transported to other subcellular membranes through complex and incompletely understood mechanisms (Thelen & Zoncu, 2017; Luo *et al.*, 2020). Lysosomal accumulation of other lipids can cause a list of lysosomal storage diseases, such as Gaucher disease (Pastores & Hughes, 1993), Batten disease (Laqtom *et al.*, 2022), and Wolman disease (Menon *et al.*, 2022). Neuronal ceroid-lipofuscinosis gene 3 (CLN3/Battenin), which is mutated in some forms of Batten disease, was recently found to mediate lysosomal egress of glycerophosphodiester, products produced from lysosomal glycerophospholipid catabolism (Laqtom *et al.*, 2022). Such lipid remodeling by CLN3 is likely linked to its impact on lysosomal membrane tubulation and vesicle trafficking (Calcagni *et al.*, 2023).

Lysosomal membranes are subject to additional dynamic regulation involving RAB7, a small GTP-binding protein, and a marker for late endosomes and lysosomes (Friedman *et al.*, 2013). Lysosomes appear to constantly associate with the ER membrane through multiple RAB7-driven tethering complexes. RAB7 is a major regulator of lysosomal fusion and transportation through its interaction with different effector proteins, such as RAB7-interacting lysosomal protein (RILP) (Jordens *et al.*, 2001), FYVE and coiled-coil domain-containing protein 1 (FYCO1) (Pankiv *et al.*, 2010), Pleckstrin homology domain-containing protein family M member 1 (PLEKHM1) (Tabata *et al.*, 2010), and protrudin (Raiborg *et al.*, 2015). Parkin, the most commonly mutated E3 ubiquitin ligase in early onset Parkinson's disease (PD), can stabilize active RAB7, which in turn promotes mitochondria-lysosome contacts for appropriate amino acid metabolism (Peng *et al.*, 2023). Additional RAB proteins particularly RAB27A mediate the peripheral trafficking of lysosomes which then undergo Ca^{2+} -dependent exocytosis upon fusion with the plasma membrane mediated by SNARE complexes (Blott & Griffiths, 2002; Tolmachova *et al.*, 2004; Tancini *et al.*, 2020).

RAB7 also interacts with multiple lipid transporters such as ORP1L (Johansson *et al.*, 2005; Boutry & Kim, 2021), PDZD8 (Guillén-Samander *et al.*, 2019; Elbaz-Alon *et al.*, 2020; Shirane *et al.*, 2020; Khan *et al.*, 2021; Gao *et al.*, 2022), and VPS13C (Hancock-Cerutti *et al.*, 2022) at the ER-lysosome contact sites and thus controls lysosomal lipid remodeling. ORP1L appears to regulate cholesterol transfer at ER-lysosome contact sites (Zhao & Ridgway, 2017), whereas PDZD8 and VPS13C are found to transfer diverse lipids with relatively weak selectivity (Kumar *et al.*, 2018; Shirane *et al.*, 2020; Gao *et al.*, 2022). Upon overexpression of a constitutively active RAB7 mutant, PDZD8 is heavily enriched at ER-lysosome contacts

(Guillén-Samander *et al*, 2019; Elbaz-Alon *et al*, 2020; Shirane *et al*, 2020; Khan *et al*, 2021; Gao *et al*, 2022). In humans, loss of PDZD8 causes a neurodevelopmental disorder that is accompanied by intellectual disability (Al-Amri *et al*, 2022). In mice and flies, homozygous deletion of PDZD8 causes deficits of long-term memory, brain structure, and synaptic plasticity (Al-Amri *et al*, 2022). VPS13C is proposed as a large-scale lipid transporter (Kumar *et al*, 2018), mutations of which are associated with the early onset PD (Lesage *et al*, 2016). VPS13C knockout in HeLa cells causes profound lysosomal expansion and broad changes in lysosomal lipid homeostasis (Hancock-Cerutti *et al*, 2022). Of note is the marked lysosomal accumulation of di-22:6-BMP, a lipid biomarker that is also seen to accumulate in patients with the G2019S mutation in the Parkinson's disease-related leucine-rich repeat kinase 2 (LRRK2) gene (Hancock-Cerutti *et al*, 2022). LRRK2, another Parkinson's disease risk gene, is reported to drive endolysosomal membrane tubulation and vesicle budding (Bonet-Ponce *et al*, 2020), which may connect with lysosomal repair (Herbst *et al*, 2020). Interestingly, the fission of lysosomal-derived tubules in this pathway also appears to happen at ER–lysosome contact sites (Bonet-Ponce & Cookson, 2022). Thus, lipid remodeling at lysosomal membrane contact sites plays essential roles in maintaining lysosomal activity.

Lysosomal quality control

Lysosomal quality control provides another, complementary strategy to preserve and maintain lysosomal activity in response to lysosomal stress or damage. At least four distinct lysosomal quality control pathways are described to date that can regenerate, remove, and repair dysfunctional lysosomes (Fig 1) (Zoncu & Perera, 2022; Yang & Tan, 2023). The transcription factor EB (TFEB)/TFE3 pathway mediates the synthesis of new lysosomes through transcriptional upregulation of lysosomal genes (Sardiello *et al*, 2009; Martina *et al*, 2014). In lysophagy, severely damaged lysosomes are captured by macroautophagy and delivered to healthy lysosomes for destruction (Hung *et al*, 2013; Maejima *et al*, 2013). The ESCRT complex, which has established roles in membrane sealing in many subcellular processes, also directly repairs lysosomal membrane when there is mild damage (Radulovic *et al*, 2018; Skowrya *et al*, 2018). We recently described an additional direct and rapid lysosomal membrane repair pathway known as PITT, for phosphoinositide-initiated membrane tethering and lipid transport (Radulovic *et al*, 2022; Tan & Finkel, 2022). The PITT pathway directly repairs lysosomal membrane pores through extensive ER–lysosome membrane contacts with subsequent ER-to-lysosome lipid transfer (Tan & Finkel, 2022). Dysfunction in lysosomal quality control pathways is known to associate with premature aging and age-related diseases. For example, TFEB expression declines with age, and restoring TFEB levels can delay senescence in various models (Wang *et al*, 2017, 2021; Zhang *et al*, 2019; Zheng *et al*, 2019). Genetic deletion of PI4K2A, the master enzyme that initiates the PITT pathway, causes lysosomal lipofuscin accumulation (a hallmark of senescence), premature aging, and neurodegeneration in human patients and mouse models (Simons *et al*, 2009; Alkhater *et al*, 2018; Dafsari *et al*, 2022).

As the first discovered lysosomal quality control mechanism, the mechanism and functions of the TFEB/TFE3 pathway have been extensively studied (Martini-Stoica *et al*, 2016; Napolitano & Ballabio, 2016; Raben & Puertollano, 2016; Yang & Tan, 2023).

TFEB/TFE3 are activated through dephosphorylation and subsequent nuclear translocation, leading to transcriptional upregulation of macroautophagy and lysosomal biogenesis. A key mechanism that controls TFEB/TFE3 subcellular localization is through mTORC1-mediated TFEB/TFE3 phosphorylation that sequesters these factors in the cytosol (Martina *et al*, 2012, 2014; Roczniak-Ferguson *et al*, 2012; Settembre *et al*, 2012). Remarkably, unlike mTORC1 canonical anabolic substrates, TFEB/TFE3 phosphorylation by mTORC1 uniquely requires folliculin-mediated activation of Rag C/D GTPases, which promotes TFEB/TFE3 recruitment for mTORC1-mediated phosphorylation (Napolitano *et al*, 2020, 2022; Jansen *et al*, 2022; Li *et al*, 2022; Cui *et al*, 2023). As such, various types of lysosomal stress activate TFEB via selective inhibition of folliculin complex or Rag C/D, without affecting canonical mTORC1 signaling (Kumar *et al*, 2020; Nakamura *et al*, 2020; Goodwin *et al*, 2021). This might explain why both TFEB and mTORC1 are simultaneously hyperactivated in senescence (see below).

In summary, the LYPAS is dynamically regulated to achieve optimal substrate processing activity for normal cellular physiology and to respond and adapt to various cellular stresses. Different LYPAS components appear to cross-talk with each other. For example, lifespan extension by methionine restriction requires autophagy-dependent lysosomal acidification (Ruckenstuhl *et al*, 2014). Increased lysosomal substrate delivery is expected to trigger lysosomal stress that in turn activates multiple lysosomal quality control pathways, membrane remodeling, and lysosomal exocytosis through Ca^{2+} signaling (Dong *et al*, 2009; Samie *et al*, 2013; Medina *et al*, 2015; Skowrya *et al*, 2018; Goodwin *et al*, 2021; Tan & Finkel, 2022).

Connecting cellular stress to LYPAS upregulation

As an essential cellular quality control system, LYPAS is upregulated by diverse types of cellular insults. Upregulation of the LYPAS includes an apparent increase in lysosomal biogenesis and substrate delivery pathways such as macroautophagy, CMA, and microautophagy, which would require simultaneous upregulation of all other LYPAS components that maintain the quality, integrity, and activity of an expanded lysosomal population (Fig 1). Depending on the level of cellular damages, cellular stresses may either be resolved or otherwise trigger senescence and/or cell death. Senescence inducers typically stimulate higher lysosomal degradative activity including substrates delivered by endocytosis, macroautophagy, and CMA (Ha *et al*, 2021; Rovira *et al*, 2022). Senescence induction by telomere shortening stimulates augmented expression of LAMP1, the autophagy proteins ATG5 and ATG12, increased LC3 lipidation, and a simultaneous reduction in the level of the autophagy adaptor p62 (Zlotorynski, 2019). Although overall LAMP2A levels are reduced with age (Cuervo & Dice, 2000a), in multiple *in vitro* models, LAMP2A expression is dramatically increased after senescence induction (Rovira *et al*, 2022), consistent with higher CMA activity in these senescent cells. Due to their higher lysosomal degradation activity (Basisty *et al*, 2020), senescent cells are also likely more vulnerable to lysosomal inhibition and therefore more dependent on rapid lysosomal repair pathways (e.g. ESCRT/PITT pathway) that ensure high lysosomal activity.

Three major mechanisms appear to connect diverse cellular stresses to LYPAS upregulation: mTORC1 signaling, Ca^{2+} signaling, and the cyclic GMP–AMP (cGAMP) synthase (cGAS)/stimulator of interferon genes (STING) pathway (Fig 2). Inactivation of mTORC1 signaling is a key mechanism in stress-induced LYPAS upregulation (Fig 2). The mTORC1 complex, normally activated on lysosomes, is often inhibited by cell stress, such as nutrient starvation, lysosomal perturbation, DNA damage, hypoxia, oxidative stress, and ER stress (Budakov & Karin, 2008; Qin *et al*, 2010; Sengupta *et al*, 2010; Aramburu *et al*, 2014; Heberle *et al*, 2015). Since mTORC1 is a major regulator of macroautophagy and TFEB, its deactivation triggers both macroautophagy and lysosomal biogenesis. Recently, mTORC1 inhibition has also been reported to stimulate V-ATPase assembly, leading to improved lysosomal acidification and hydrolytic degradation (Ratto *et al*, 2022). Like mTORC1, TFEB is also activated by diverse cellular stresses (Raben & Puertollano, 2016; Yang & Tan, 2023), but not all of

these conditions cause mTORC1 deactivation. For example, cellular stresses that trigger ATG8 lipidation (also known as ATG8ylation (Kumar *et al*, 2021), conjugation of ATG8 to single membranes or CASM (Durgan & Florey, 2022)) or cellular energy crisis with AMPK signaling can activate TFEB without affecting conical mTORC1 signaling (Goodwin *et al*, 2021; Malik *et al*, 2023). In such contexts, disruption of RagC/D-mediated TFEB loading to mTORC1 is emerging as a major mechanism underlying TFEB activation (Goodwin *et al*, 2021; Malik *et al*, 2023). Even when canonical mTORC1 signaling is hyperactivated such as in senescent cells (Herranz *et al*, 2015; Laberge *et al*, 2015; Carroll *et al*, 2017), TFEB can still be highly activated for profound lysosomal expansion (Rovira *et al*, 2022; Curnock *et al*, 2023). Thus, downstream of various cellular stresses, modification of mTORC1 signaling leads to upregulation of multiple aspects of LYPAS including macroautophagy, lysosomal biogenesis, and V-ATPase-mediated lysosomal acidification.

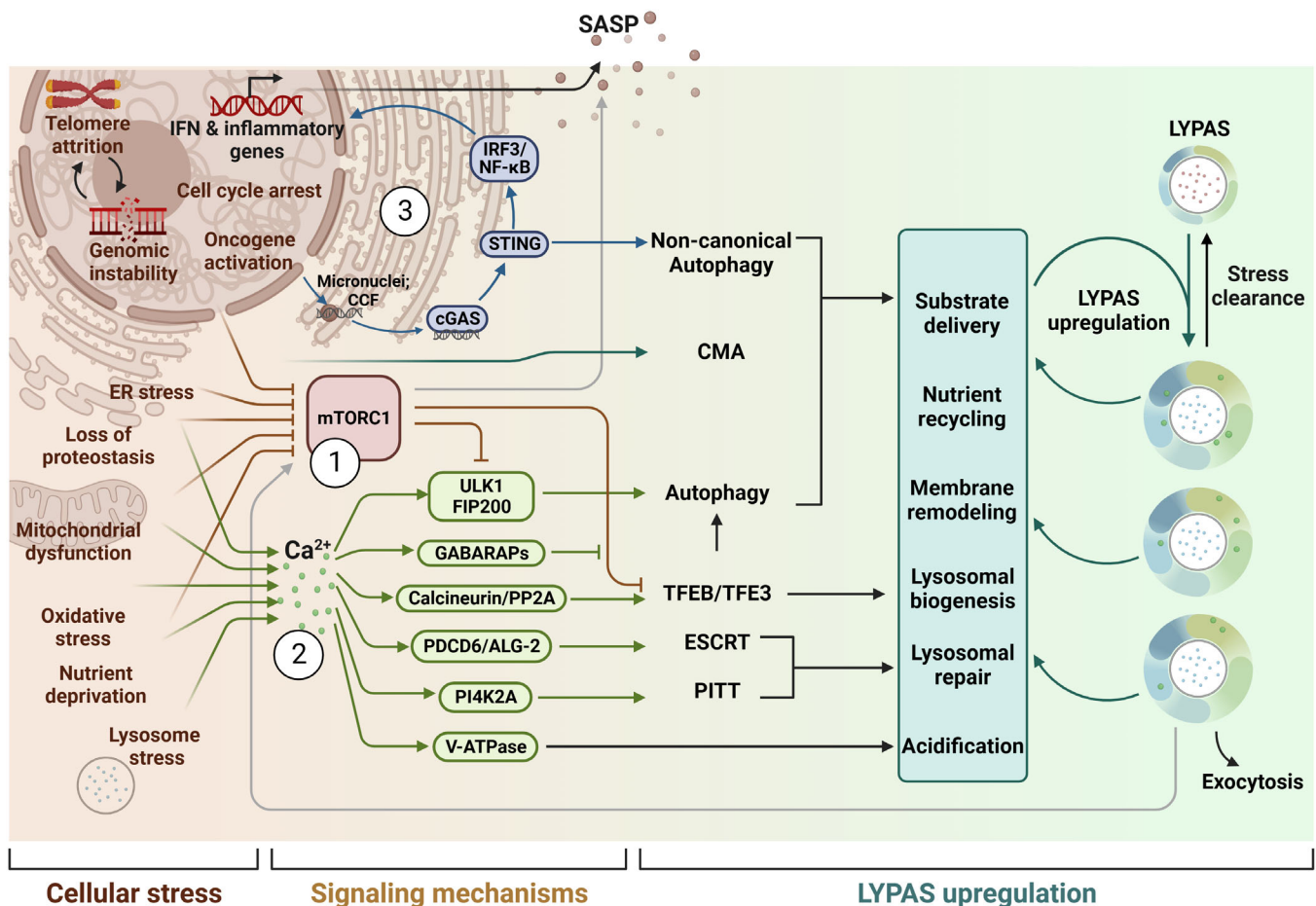


Figure 2. Diverse cellular stresses trigger LYPAS upregulation.

The components of the LYPAS are upregulated in response to many different types of cellular stresses. (1) Regulation of mTORC1 activity by various stressors modulates the level of substrate delivery and lysosomal capacity through autophagy and TFEB activation. (2) Ca^{2+} signaling provides another common stress-induced signal that activates many downstream effectors in autophagy initiation, TFEB activation, lysosomal acidification, and lysosomal repair pathways including ESCRT and PITT. (3) Cytosolic DNA exposure is a common stress response signal that activates non-canonical autophagy as well as SASP through the cGAS/STING pathway of DNA sensing. CMA, one of the lysosomal substrate delivery mechanisms, is also activated as part of a stress response potentially by protein misfolding. Graphics were created with BioRender.com.

Ca^{2+} signaling is emerging as a universal mechanism for LYPAS upregulation in response to numerous cellular stresses (Fig 2). In resting conditions, cytosolic Ca^{2+} levels are kept below 100 nM, whereas almost all cellular insults, through diverse mechanisms, provoke cytosolic Ca^{2+} concentration either locally or throughout the cytoplasm (Hu *et al.*, 2022b). Ca^{2+} signaling is a crucial mechanism for autophagy initiation, which has been recently found to induce the phase separation and thus activation of the autophagy initiation factor FIP200 (Zheng *et al.*, 2022a). Ca^{2+} signaling is also critical for TFEB/TFE3 activation in lysosomal biogenesis and transcriptional autophagy upregulation (Yang & Tan, 2023). Lysosomal Ca^{2+} release by TRPML1 activates TFEB through V-ATPase-dependent, lysosomal ATG8 lipidation, and subsequent sequestration of the folliculin complex, an activator of RagC/D required for mTORC1-mediated TFEB phosphorylation and suppression (Kumar *et al.*, 2020; Nakamura *et al.*, 2020; Goodwin *et al.*, 2021). In other contexts, Ca^{2+} signaling appears to activate TFEB phosphatases such as calcineurin or PP2A (Medina *et al.*, 2015; Hasegawa *et al.*, 2022). Ca^{2+} signaling additionally initiates other lysosomal quality control pathways, such as the ESCRT and the PITT pathways for rapid lysosomal repair (Radulovic *et al.*, 2018; Skowrya *et al.*, 2018; Tan & Finkel, 2022). Thus, cellular stressors triggering Ca^{2+} signaling, directly or indirectly linked to lysosomal stress, all have the potential to activate lysosomal biogenesis, quality control, autophagy, and acidification.

A crucial pathway activated by more severe stressors such as genomic instability is the cyclic GMP–AMP (cGAMP) synthase (cGAS)/stimulator of interferon genes (STING) innate immune pathway (Ishikawa & Barber, 2008; Zhong *et al.*, 2008; Ishikawa *et al.*, 2009; Sun *et al.*, 2009) which has a highly conserved function in autophagy upregulation (Fig 2) (Moretti *et al.*, 2017; Gui *et al.*, 2019; Liu *et al.*, 2019). The initiating enzyme cGAS is activated upon the sensing of cytosolic DNA (Sun *et al.*, 2013) which can occur in response to various cellular stresses (Harding *et al.*, 2017; Kang *et al.*, 2018; Nassour *et al.*, 2019; Lv *et al.*, 2022). Upon DNA binding, cGAS produces the second messenger cGAMP that binds to and activates the signaling adaptor STING for autophagy induction (Sun *et al.*, 2013; Wu *et al.*, 2013; Gui *et al.*, 2019). STING-induced autophagy appears to involve the direct recruitment of WIPI2 independently of other upstream autophagy initiation factors such as ULK1, Beclin 1, ATG14, and VPS34 (Gui *et al.*, 2019; Liu *et al.*, 2019; Wan *et al.*, 2023). While the production of cytosolic self-DNA stimulates the cGAS/STING pathway (Dou *et al.*, 2017; Glück *et al.*, 2017; Yang *et al.*, 2017), these DNA fragments are cleared by LYPAS through STING-dependent, non-canonical autophagosome formation. Even after senescence entrance, cytoplasmic chromatin fragments (CCFs) and the nuclear lamina protein lamin B1 are continuously processed by the LYPAS, causing progressive depletion of histones (Ivanov *et al.*, 2013; Dou *et al.*, 2015).

Beside autophagy induction, the cGAS/STING pathway is a critical contributor to SASP, a key feature of cellular senescence, through transcriptional upregulation of interferons and pro-inflammatory cytokines through Interferon regulatory factor 3 (IRF3) and NF- κ B (nuclear factor of kappa light polypeptide gene enhancer in B-cells) signaling (Fig 2) (Harding *et al.*, 2017; Kang *et al.*, 2018; Nassour *et al.*, 2019; Lv *et al.*, 2022). Common forms of cytosolic cGAS-activating self-DNA include micronuclei resulting from erroneous chromosome segregation and CCFs formed during cellular

insults such as DNA damage, oncogene activation, and replicative exhaustion (Miller *et al.*, 2021). Telomere shortening, a common marker of aging, triggers senescence by DNA damage-induced CCF formation and subsequent cGAS activation (Takai *et al.*, 2003; Abd-salaam *et al.*, 2020; Lv *et al.*, 2022). Of note, dysfunctional mitochondria, which accumulate in senescence, can initiate retrograde signaling to promote CCF formation, activating cGAS-mediated SASP (Vizioli *et al.*, 2020). Cytosolic self-DNA might also be derived from dysfunctional mitochondria themselves. For example, telomere stress might communicate with mitochondria causing mitochondrial DNA leakage, which activates cGAS for subsequent SASP induction (Kang *et al.*, 2018; Lv *et al.*, 2022). More recently, cGAS has been found to also detect cytosolic DNA of de-repressed endogenous retrovirus HERVK in senescent human cells, which is not observed in early passage non-senescent cells (Liu *et al.*, 2023). Upon cGAMP binding, STING traffics from the ER to perinuclear Golgi-derived vesicles, where downstream TBK1/IRF3 and NF- κ B signaling are activated for the secretion of type I interferons and pro-inflammatory cytokines (Harding *et al.*, 2017; Kang *et al.*, 2018; Tan *et al.*, 2018; Nassour *et al.*, 2019; Lv *et al.*, 2022).

Senescence-inducing stresses also instigate abnormal mTORC1 signaling on hyperactivated lysosomes as another major mediator of SASP (Herranz *et al.*, 2015; Laberge *et al.*, 2015; Carroll *et al.*, 2017), which reinforces cGAS/STING-mediated secretion (Laberge *et al.*, 2015) (Fig 2). Elevated lysosomal substrate delivery and hydrolytic breakdown generate higher levels of intracellular amino acids, which, in turn, supports constant activation of the mTORC1 nutrient-sensing complex (Herranz *et al.*, 2015; Laberge *et al.*, 2015; Carroll *et al.*, 2017), consistent with increased lysosomal association of mTORC1 in senescence (Narita *et al.*, 2011; Rovira *et al.*, 2022). Multiple senescence inducers such as DNA damage, oncogene activation, and replicative exhaustion also trigger abnormal mTORC1 signaling through lysosomal accumulation of cholesterol mediated by a lysosomal cholesterol importer ABCA1 (Roh *et al.*, 2023) and potentially lysosomal stress-induced ER-to-lysosome cholesterol transfer by the PITT pathway (Tan & Finkel, 2022). Thus, increased lysosomal degradation and constitutive mTORC1 activation (Herranz *et al.*, 2015; Laberge *et al.*, 2015) underlie accelerated protein turnover, synthesis, and secretion in senescent cells. Of note, abnormal mTORC1 signaling in senescent cells might inhibit macroautophagy (Ott *et al.*, 2016b), causing damage accumulation.

Lysosomal processing and adaptation system upregulation in senescence includes enhanced lysosomal exocytosis, which partially contributes to SASP. Proteins secreted by SASP include a myriad of cytokines, chemokines, growth factors, and proteases, altogether contributing to chronic inflammation and tissue damage in various age-related pathologies (Muñoz-Espín & Serrano, 2014). Upregulated age-dependent serum SASP factors include a list of lysosomal proteins such as cathepsins B, D, Z, metalloproteinase MMP2, and gelsolin (Basisty *et al.*, 2020; Rovira *et al.*, 2022). Consistent with increased lysosomal secretion via exocytosis, senescent cells have much higher levels of the lysosomal transmembrane proteins LAMP1 and LAMP2 on their plasma membrane (Rovira *et al.*, 2022). Knockdown of RAB27A, a key small GTPase involved in exocytosis, dramatically diminished the secretion of lysosomal enzymes from senescent cells (Rovira *et al.*, 2022). Given the robust upregulation of autophagy in senescence, it seems likely that secretory autophagy

(Cadwell & Debnath, 2018; New & Thomas, 2019) might also contribute to SASP.

In summary, in response to various forms of cellular stress including those that trigger cellular senescence, the LYPAS is expanded for higher substrate processing activity in order to help resolve the stress. The underlying mechanisms include mTORC1 regulation, Ca²⁺ signaling, and cGAS/STING activation, leading to broad LYPAS upregulation such as macroautophagy initiation, lysosomal biogenesis, acidification, and exocytosis. When the stresses are more severe, these LYPAS-expanding pathways also contribute to senescence induction through transcriptional boosting of SASP and lysosomal exocytosis.

LYPAS overload in senescence and cell death

Lysosomes are essential for proteostasis, organelle quality control, and the resolution of cellular stresses. Senescence, typically induced by unresolvable cellular stress, is a cellular state characterized by a permanent cell cycle arrest, increased genomic instability, decreased cellular homeostasis, and accumulation of dysfunctional organelles such as mitochondria. As discussed above, senescence-inducing factors broadly upregulate the LYPAS. However, despite elevated lysosomal biogenesis and accelerated lysosomal substrate delivery, senescent cells paradoxically exhibit lysosomal accumulation of lipofuscin and senescence-associated β-galactosidase, as well as lysosomal pH neutralization (Hernandez-Segura *et al.*, 2018; Huang *et al.*, 2022). Indeed, lipofuscin and β-galactosidase are used as markers for cellular senescence in many situations (Dimri *et al.*, 1995; Kurz *et al.*, 2000; Debaq-Chainiaux *et al.*, 2009). Lysosomal fractions from proliferating and senescent cells *in vitro* showed similar total protein levels and similar activity of lysosomal enzymes (Rovira *et al.*, 2022). Thus, a major reason for lipofuscin accumulation might be increased delivery of substrates to lysosomes instead of a decline in lysosomal degradation activity. The increased

amounts of lysosomal mass and lipofuscin storage over time might reflect a deepening of the senescent state, which could ultimately further compromise lysosomal integrity and activity (Pan *et al.*, 2021).

The current understanding of the relationship between senescence and the LYPAS can be summarized into four aspects: (i) senescent cells exhibit lysosomal expansion and dysfunction; (ii) most cellular insults including senescence-inducing stimuli upregulate LYPAS activity; (iii) perturbing LYPAS activity triggers senescence (accompanied by accumulation of oxidative damage, dysfunctional mitochondria, and genomic instability) or cell death, depending on how LYPAS is perturbed; (iv) boosting LYPAS capacity suppresses senescence and cell death, thereby maintaining or rejuvenating cellular functions.

The interconnected observations above support a lysosomal model of senescence/cell death. This model suggests that with damage, a cell fate decision between senescence and cell death integrates the substrate processing capacity of a cell's LYPAS and the overall cellular stress/damage level (Fig 3A): (i) Senescence is not initiated when the level of cellular stress and damage remains within the handling capacity of the LYPAS; (ii) senescence is induced when lysosomes are continuously overloaded such that an imbalance between substrate loading and saturated lysosomal digestion is achieved with, most importantly, gradual accumulation of more damaged materials; (iii) senescence maintenance likely requires a certain high level of *lysosomal* activity, which handles continuous and ongoing cellular stress and might also reinforce senescence by mTORC1-mediated SASP; (iv) the initial phase of senescence might be reversible upon resolution of the stress, while pharmacologically or genetically inhibiting LYPAS activity might accelerate senescence or cell death, depending on the level of stress (García-Prat *et al.*, 2016; Fujimaki *et al.*, 2019).

Based on this lysosomal model of senescence/cell death, either increased cellular stress or compromised LYPAS activity can cause senescence entrance due to insufficient capacity in handling

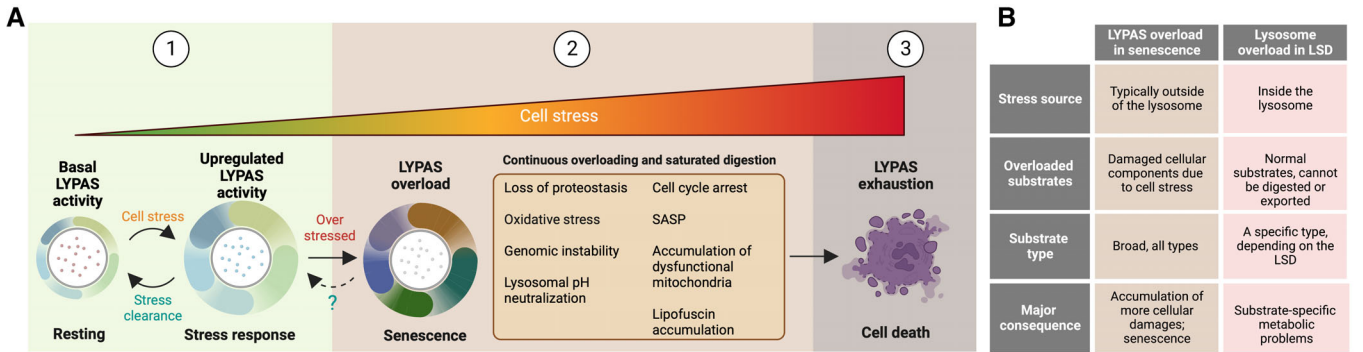


Figure 3. A lysosomal-centric model of senescence and cell death—LYPAS overload in senescence induction.

(A) Cellular stress triggers LYPAS upregulation to process an increase in damaged macromolecules caused by stress. (1) Cellular stresses are resolved without senescence induction when the substrate processing capacity of the upregulated LYPAS is above the substrate-producing (macromolecule-damaging) level. Upon stress clearance, the LYPAS is returned to its basal level. (2) Senescence is induced when higher levels of persistent cellular stress causes LYPAS overload which cannot be resolved effectively, causing a permanent cell cycle arrest with potential progression to LYPAS exhaustion. Persistent damage continues to cause more macromolecule damage, leading to damage accumulation and gradual dysfunction of expanded lysosomes. (3) LYPAS exhaustion breaks the equilibration in senescence and causes cell death. In some situations, the cellular stress directly causes LYPAS exhaustion and thus causes cell death without a detectable senescence stage. (B) Distinctive feature of LYPAS overload in senescence versus the lysosomal overload observed in lysosomal storage disease (LSD). Graphics were created with BioRender.com.

damaged cellular components. Besides stress-induced senescence, dysfunctional lysosomal activity or autophagy has been shown to cause senescence in different models (Kang *et al*, 2011; García-Prat *et al*, 2016; Fujimaki *et al*, 2019; Dutta *et al*, 2022). It is not clear how LYPAS overload exactly triggers senescence entrance, but this likely involves the cellular accumulation of multiple abnormal materials, such as misfolded proteins and dysfunctional mitochondria, which in turn may impair DNA repair mechanisms leading to increased DNA damage (Fig 3A). Genomic instability is a key feature of senescent cells, which can contribute to cell cycle arrest and the activation of the cGAS/STING signaling for SASP upregulation.

Upon senescence induction, the maintenance of the senescent state requires continuous LYPAS activity. A recent report showed that nuclear localization of TFEB, and the related transcription factor TFE3, was a seeming universal hallmark of the senescent state (Curnock *et al*, 2023). When damaged cells on their way to senescence were subjected to knockdown of these transcription factors, they failed to undergo the transition to senescence and instead underwent cell death (Curnock *et al*, 2023). Similarly, in chemotherapy-induced senescent lymphoma cells, directly blocking lysosomal degradation by the V-ATPase inhibitor bafilomycin A1 or concanamycin A can trigger cell death (Dörr *et al*, 2013). Senescence maintenance likely requires the combined engagement of multiple branches of the LYPAS. This would encompass pre-existing lysosomes and newly synthesized ones through the TFEB/TFE3 pathway, the help of additional lysosomal quality control mechanisms (e.g. lysophagy, ESCRT, and PITT), as well as upregulated lysosomal delivery through macroautophagy, microautophagy, and chaperone-mediated autophagy (Fig 1).

It is important to note that the LYPAS overload in senescence appears to be distinct from lysosome overload in LSD, as their causes and consequences are rather different (Fig 3B). In senescence, the overload is due to stress-induced, broad damages of many other cellular components outside of the lysosome, which saturates not only lysosomal digestion but also all other aspects of LYPAS including substrate delivery. Under such conditions, cells will gradually accumulate more and more cellular damage over time, a key feature of senescent cells. However, in LSD, typically only one type of substrates build up due to the genetic loss of function of a specific lysosomal enzyme (Platt *et al*, 2018), which only indirectly affects lysosomal delivery and digestion of other substrates. Thus, different from LYPAS overload in senescence, lysosomal overload in LSD often causes substrate-specific metabolic problems without apparent senescent or aging phenotypes.

The removal of senescent cells occurs through immune-mediated pathways and presumably through cell-autonomous cell death pathways. We posit that the death of senescent cells is likely connected to lysosomal exhaustion, which we refer to as a state of persistent lysosomal damage exceeding the reparative capacity of the LYPAS, primarily due to continuous overloading and lysosomal accumulation of undigested materials (Gómez-Sintes *et al*, 2016; Wang *et al*, 2018) (Fig 3). As such, in aged and diseased tissues, the expansion of dysfunctional lysosomes might reflect different stages of the senescence/cell death continuum. A lack of current markers to experimentally distinguish different stages of lysosomal dysfunction renders it difficult to delineate the precise relationship between lysosomal activity and disease progression.

Augmenting LYPAS capacity for disease treatment and healthy aging

Given the broad spectrum of LYPAS substrates, improving LYPAS capacity represents a promising approach for healthy aging and the treatment of age-related diseases. The overall substrate processing capacity and the maintenance efficiency of the LYPAS decrease with age, as exemplified by less macroautophagy (Leidal *et al*, 2018), CMA (Cuervo & Dice, 2000a), microautophagy (Krause *et al*, 2022), and TFEB expression (Wang *et al*, 2021). Considering that increased cellular stress in aging requires higher LYPAS activity for stress resolution, the age-dependent decrease in LYPAS capacity might inevitably trigger senescence and/or cell death in age-related conditions (Fig 4A). Lower than normal LYPAS capacity would trigger earlier disease onset with LSDs as an extreme example, whereas milder lysosomal deficits exhibit a slower, age-dependent progression. In contrast, improving the LYPAS capacity would be expected to, and in some cases has been demonstrated to, delay or reverse cellular senescence, improve immune clearance of senescent cells, alleviate age-related pathology, and promote healthy aging (Fig 4B). Below we discuss strategies to improve the LYPAS capacity by targeting different LYPAS components (Fig 4C).

Boosting lysosomal biogenesis and substrate delivery is perhaps the most straightforward strategy to improve LYPAS capacity. TFEB is extensively studied as a therapeutic target for age-related diseases, as it can mediate both lysosomal expansion and autophagy-dependent substrate delivery. Besides autophagy, TFEB also appears to enhance endocytosis-mediated lysosomal substrate delivery which might help clear extracellular protein aggregates (Martini-Stoica *et al*, 2018). Although TFEB is activated upon senescence induction by different stressors (Rovira *et al*, 2022; Curnock *et al*, 2023), during aging the levels of TFEB in various tissues are indeed decreased (Zheng *et al*, 2018; Zhang *et al*, 2019; Wang *et al*, 2021). Restoring TFEB expression has been shown to protect cells from senescence and ameliorate senescence-related pathology *in vitro* and *in vivo* (Zhang *et al*, 2019; Zheng *et al*, 2019, 2018; Wang *et al*, 2021). The Parkinson's disease-related gene LRRK2 is recently found to suppress TFEB activation and lysosomal biogenesis, and the genetic deletion or pharmacological inhibition of LRRK2 restored lysosomal abundance and proteolytic activity (Yadavalli & Ferguson, 2023). Spermidine, an endogenous metabolite of polyamine, is reduced in the elderly, whereas its supplementation restores TFEB expression and reverses immune senescence (Zhang *et al*, 2019). Mechanistically, spermidine post-translationally modifies eukaryotic translation initiation factor 5A (eIF5A), allowing more efficient translation of TFEB (Zhang *et al*, 2019). Additional promising small molecule activators of the TFEB pathway have been discovered or developed in recent years, some of which are under clinical investigation (Wang *et al*, 2017; Silvestrini *et al*, 2018; Liang, 2020; Liu *et al*, 2021; Yang *et al*, 2022; preprint: Yoon *et al*, 2022). The mTOR inhibitor rapamycin is well-known for its impact on healthy aging and longevity promotion in multiple model organisms (Johnson *et al*, 2015). mTORC1 is a master regulator of autophagy and a driver of SASP (Herranz *et al*, 2015; Laberge *et al*, 2015; Carroll *et al*, 2017; Liu & Sabatini, 2020). Although it does not seem to activate TFEB (Settembre *et al*, 2011), rapamycin potentially stimulates macroautophagy-mediated lysosomal substrate delivery (Carmona-Gutierrez *et al*, 2016) and strongly suppresses

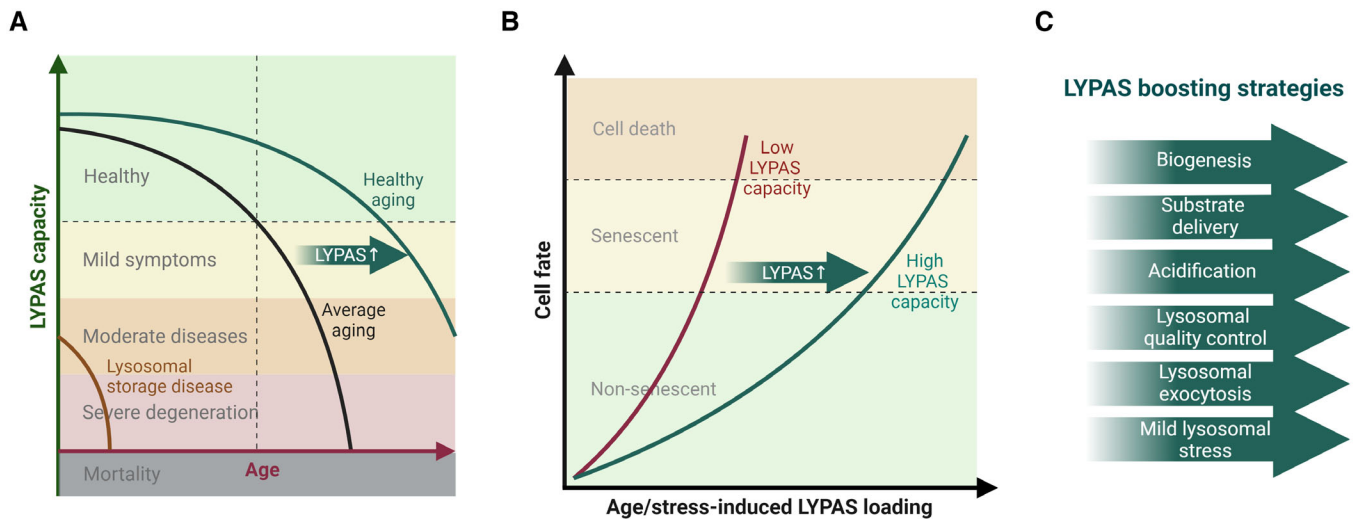


Figure 4. Boosting the capacity of the lysosomal processing and adaptation system for disease treatment and healthy aging.

(A) An illustration of the relationship between LYPAS capacity and age in different contexts. (B) Schematic graph showing the relationship between cell fate and stress-related LYPAS substrate loading in high or low LYPAS capacity individuals. (C) Strategies to potentially boost LYPAS capacity for treating disease and promoting healthy aging. Green arrows indicate strategies that can boost the LYPAS capacity, thereby shifting the curves in (A) and (B) rightward. Graphics were created with [BioRender.com](https://www.biorender.com).

SASP (Herranz *et al*, 2015). As such, rapamycin is known to inhibit senescence or senescence-related pathology in multiple cell types and tissues including epithelial stem cells (Herranz *et al*, 2015), hematopoietic stem cells (Chen *et al*, 2009), aging hearts (Dai *et al*, 2014), neurons (Singh *et al*, 2019), human skin cells (Chung *et al*, 2019), and bone cells (An *et al*, 2020, 2017). Lysosomal substrate delivery can be alternatively stimulated by CMA activators, such as geldanamycin (GA), 6-aminomycinamide (6-AN), and atypical retinoid 7 (AR7) (Finn *et al*, 2005; Anguiano *et al*, 2013). A recently generated more potent derivative of AR7 (CA77.1) has shown promising *in vivo* activity in selectively activating CMA, thereby ameliorating pathology in a mouse model of Alzheimer's disease (Bourdenx *et al*, 2021). Similar to CMA activators, inducers of microautophagy can also boost LYPAS capacity, since this pathway serves as another key substrate delivery route upregulated by cellular stress (Goodwin *et al*, 2017; Mejlvang *et al*, 2018; Olsvik *et al*, 2019; Wang *et al*, 2023).

Improving lysosomal acidification amplifies the LYPAS capacity by keeping lysosomal hydrolytic enzymes at their optimal activity. Overexpression of V-ATPase subunits is sufficient to improve lysosomal acidification and provide health benefits, such as motor activity improvements in Parkinson's disease mouse models (Jin *et al*, 2012) and lifespan extension in yeasts (Ruckenstuhl *et al*, 2014). Alternatively, overexpression or activation of TFEB can promote lysosomal acidification by transcriptional upregulation of lysosomal genes including V-ATPase subunits (Palmieri *et al*, 2011; Zhang *et al*, 2015; Bouché *et al*, 2016). Interestingly, mTORC1 inhibition has been recently reported to stimulate V-ATPase assembly, improving lysosomal acidification and hydrolytic degradation (Ratto *et al*, 2022), which might also contribute to the benefits of rapamycin, together with autophagy upregulation and potential TFEB activation. A recent study identified a fragment of amyloid

precursor protein (APP) as a tonic inhibitor of V-ATPase assembly in mouse models of Down syndrome and AD (Im *et al*, 2023). As such, reducing the levels of the APP fragment triggered V-ATPase assembly and increased lysosomal acidification (Im *et al*, 2023). Agonists of the lysosomal Ca^{2+} channel TRPML1 stimulate lysosomal acidification (Bae *et al*, 2014) likely through direct V-ATPase activation by Ca^{2+} -bound calmodulin (Peters *et al*, 2001) or TFEB activation (Palmieri *et al*, 2011; Zhang *et al*, 2015; Bouché *et al*, 2016), and have been shown to benefit lysosomal-related disease models (Hui *et al*, 2019; Pollmanns *et al*, 2022). In contrast to TRPML1, TPCN2-mediated Ca^{2+} release appears to raise lysosomal pH (Tong *et al*, 2022). As such, a TPCN2 inhibitor tetrandrine has been found to provide benefits in an Alzheimer's disease mouse model (Tong *et al*, 2022). As CLC7 is a H^+/Cl^- antiporter that promotes lysosomal acidification (Graves *et al*, 2008), increasing the expression or lysosomal targeting of CLC7 has been shown to increase lysosomal acidification. For example, macrophage colony-stimulating factor (MCSF) or interleukin-6 stabilizes and targets CLC7 to lysosomes in microglia, leading to decreased lysosomal pH and accelerated degradation of amyloid fibrils (Majumdar *et al*, 2011). The phosphoinositide messenger PI(3,5) P_2 regulates lysosomal pH by regulating multiple ion channels including the V-ATPase (Li *et al*, 2014; Banerjee & Kane, 2020), CLC7 (Leray *et al*, 2022; Hilton & Mindell, 2023), and TRPML1 (Fine *et al*, 2018; Gan *et al*, 2022). Inhibition of PIKfyve is known to activate V-ATPase through CLC7 activation and thus trigger lysosomal hyper-acidification (Leray *et al*, 2022; Hilton & Mindell, 2023). Although the loss of function of PIKfyve can cause neurodegenerative disease, partial inhibition of PIKfyve might provide benefits for certain diseases such as amyotrophic lateral sclerosis (ALS) for which there have been few effective therapeutic options (see below).

Given that any expanded system requires more extensive monitoring and maintenance, preserving lysosomal integrity through improved lysosomal quality control pathways might be particularly important, allowing for maximal expansion of LYPAS capacity and thus inhibiting or delaying senescence (Kirkegaard *et al.*, 2010; García-Prat *et al.*, 2016; Fujimaki *et al.*, 2019; Peng *et al.*, 2019). TFEB activators described above should also promote lysosomal quality. Since most lysosomal quality control pathways including TFEB, ESCRT, and PITT can be activated by lysosomal Ca^{2+} signaling (Fig 2), stimulating lysosomal Ca^{2+} release might improve lysosomal quality. For example, TRPML1 agonists are known to activate both TFEB (Goodwin *et al.*, 2021) and PITT (Tan & Finkel, 2022) in the absence of lysosomal membrane damage. Thus, pharmacological activation of lysosomal channels like TRPML1 might be a powerful approach to improve both lysosomal quantity and quality, although the long-term consequences of depleting luminal lysosomal calcium is not well understood. Additionally, agonists of PI4K2A, the key enzyme initiating the PITT pathway (Tan & Finkel, 2022), are expected to provide robust lysosomal protection against potential lysosomal damaging sources such as internalized tau fibril aggregates (Zhao *et al.*, 2021) commonly found in Alzheimer's disease.

Instead of direct hydrolytic degradation, the substrate processing capacity of LYPAS might be alternatively enhanced by lysosomal exocytosis. This approach directly releases the LYPAS burden of donor cells such as neurons or cardiomyocytes, allowing the participation of the immune system in substrate clearance. Since it also lowers the risk of lysosomal storage/swelling or aggregate-mediated lysosomal membrane damage, it is thus a reasonable choice in situations where the lysosomal hydrolytic capacity is limited or overwhelmed despite the upregulation of LYPAS. Lysosomal exocytosis involves RAB27A-mediated lysosomal trafficking to the plasma membrane (Blott & Griffiths, 2002) and subsequent SNARE-mediated final membrane fusion (Rao *et al.*, 2004). RAB27A is down-regulated in cystinosis (Johnson *et al.*, 2013), a lysosomal cystine storage disease due to mutations of cystinosis, the cysteine transporter (Kalatzis *et al.*, 2001). Restoring RAB27A expression in cystinosis-knockout cells rescues lysosomal peripheral trafficking and exocytosis, relieving lysosomal storage of cysteine (Johnson *et al.*, 2013). TFEB activation or TRPML1-mediated lysosomal Ca^{2+} signaling can promote lysosomal exocytosis as one of their lysosomal protective activities (LaPlante *et al.*, 2006; Dong *et al.*, 2009; Medina *et al.*, 2011; Samie *et al.*, 2013; Martina *et al.*, 2014). TRPML1 not only activates TFEB through Ca^{2+} signaling but the channel is itself a transcription target of TFEB (Palmieri *et al.*, 2011), thus establishing a positive feedback loop in this lysosomal stress response. Overexpression of TFEB can promote lysosomal substrate clearance through TRPML1-mediated exocytosis (Wang *et al.*, 2015; Xu *et al.*, 2021). Chemical agonists of TRPML1 have been shown to stimulate lysosomal exocytosis, which provides protection from NPC disease (Shen *et al.*, 2012) or α -synuclein toxicity (Tsunemi *et al.*, 2019). Similarly, low concentrations of cyclodextrin, which are cyclic oligosaccharides, can moderate lysosomal cholesterol storage in NPC disease models by triggering TRPML1-mediated lysosomal exocytosis (Chen *et al.*, 2010; Vacca *et al.*, 2019). Chemical activators of two pore channel 2 (TPC2, another Ca^{2+} channel) also stimulates lysosomal exocytosis and increases autophagy, thereby alleviating lysosomal storage in MLIV and Batten disease models (Scotto Rosato *et al.*, 2022). Therefore, stimulating lysosomal

exocytosis may be an effective approach to mitigate lysosomal substrate overload, especially for LSDs including lipid storage diseases.

Recently, inducing mild lysosomal stress is emerging as an encouraging therapeutic approach that expands the LYPAS capacity. Likely similar to the benefits of mild proteasomal or mitochondrial stress (mitohormesis) (Yun & Finkel, 2014; Schulz & Haynes, 2015), mild lysosomal perturbations have proven beneficial (lysosomosis) in many disease models. Chemicals capable of triggering lysosomal hormesis include metformin, trehalose, and PIKfyve inhibitors. Metformin, widely prescribed for type II diabetes, has been recently found to induce mild lysosomal stress that offers anti-aging benefits through the activation of adenosine monophosphate-activated protein kinase (AMPK) (Ma *et al.*, 2022). Clinically relevant concentrations of metformin binds to a subunit of γ -secretase, PEN2, which in turn forms a complex with V-ATPase to inhibit lysosomal acidification (Ma *et al.*, 2022), leading to AMPK activation through a complex of V-ATPase and AMPK (Zhang *et al.*, 2017), autophagy induction, and potentially induction of lysosomal quality control pathways.

Trehalose, a natural disaccharide, has been known as a potent autophagy stimulator that promotes clearance of many pathological proteins such as mutant huntingtin (Sarkar *et al.*, 2007), α -synuclein (Sarkar *et al.*, 2007), TDP-43 (Wang *et al.*, 2010), tau aggregates (Wang *et al.*, 2010; Schaeffer & Goedert, 2012), and β amyloid (Perucho *et al.*, 2012) through mTORC1-independent mechanisms (Sarkar *et al.*, 2007). Trehalose broadly alleviates pathology in many diseases models (Vessey *et al.*, 2022), such as LSDs (Palmieri *et al.*, 2017; Lotfi *et al.*, 2018), Parkinson's disease (Khalifeh *et al.*, 2019; Pupyshv *et al.*, 2019), atherosclerosis (Sergin *et al.*, 2017; Evans *et al.*, 2018), amyotrophic lateral sclerosis (ALS) (Zhang *et al.*, 2014), and frontotemporal dementia (Holler *et al.*, 2016). Early human clinical trials are underway (Mobini *et al.*, 2022). Recently, trehalose has been found to accumulate in lysosomes, causing lysosomal membrane permeabilization and/or a mild pH neutralization, which activates TFEB for autophagy upregulation and lysosomal biogenesis (Evans *et al.*, 2018; Rusmini *et al.*, 2019; Jeong *et al.*, 2021). It is still unclear what exact lysosomal stress is induced by trehalose and what immediate effector(s) respond to such stress to activate TFEB and other lysosomal protection pathways.

Another groups of compounds that appear to benefit the LYPAS through mild lysosomal stress are PIKfyve inhibitors. Although PIKfyve mutations can cause neurodegeneration, partial PIKfyve inhibition by small molecules, including the clinically available apilimod and YM201636, appears to ameliorate various neurodegenerative diseases (Shi *et al.*, 2018; Soares *et al.*, 2021; Hung *et al.*, 2023). PIKfyve inhibition promoted neuronal survival and relieved neurodegeneration in diverse models of ALS both *in vitro* and *in vivo*, which appears to involve, at least in part, exocytosis of aggregation-prone proteins (Shi *et al.*, 2018; Hung *et al.*, 2023). In addition, PIKfyve inhibition suppresses the trafficking of endocytosed tau seeds to lysosomes and their subsequent cytosolic spreading, with no impact on their endocytosis (Soares *et al.*, 2021). It appears that partial inhibition of PIKfyve is well tolerated (Billich, 2007) and might induce a hormetic-like, mild lysosomal stress that can potentially be beneficial to neurodegenerative disease patients. Several mechanisms might be underlying the benefits of PIKfyve inhibition in neurodegenerative diseases characterized by protein aggregation: (i) A partial inhibition of PIKfyve might increase lysosomal degradation activity by triggering

hyper-acidification of lysosomes (Leray *et al*, 2022; Hilton & Mindell, 2023); (ii) an enlarged endolysosomal lumen can hold more protein aggregates than usual with reduced risk of lysosomal membrane damage by protein aggregates, the final step of cytosol invasion by internalized aggregates; (iii) increased lysosomal membrane tension likely promotes lysosomal exocytosis to expel internalized protein aggregates; (iv) PIKfyve inhibition activates TFEB, a major regulator of transcriptional lysosomal biogenesis and autophagy upregulation (Hasegawa *et al*, 2022).

In summary, the LYPAS capacity can be improved by multiple approaches, such as lysosomal biogenesis, substrate delivery, lysosomal acidification, lysosomal quality control, lysosomal exocytosis, and broader LYPAS enhancement through mild hormetic lysosomal stress. Lysosomal Ca^{2+} signaling likely interconnects many therapeutic approaches and various LYPAS components for the final augmentation of LYPAS capacity in substrate processing.

Concluding remarks

Lysosomal dysfunction is commonly found in pathological conditions and is receiving increased attention in the biology of aging and in the pathogenesis of age-related diseases. The exact roles of lysosomal dysfunction in senescence, cell death, or disease progression have been puzzling for a long time, partially due to the difficulty of studying these processes *in vivo*. Considering the close relationship between lysosomal activity and disease severity in numerous models and from human subjects, further investigations delineating how lysosomes participate in cellular stress responses, senescence induction and maintenance, and cell death decisions are needed. To facilitate such investigations, we introduced here the concept of the lysosomal processing and adaptation system or LYPAS. We discussed the major components of this system that maintains the substrate processing capacity of lysosomes (Fig 1) and how LYPAS is boosted in response to diverse cellular stresses including those stimulating senescence or cell death (Fig 2). We discussed the concept of lysosomal overload as a point of no return in senescence induction and lysosomal exhaustion as a potential decision point whereby a senescent cell proceeds to cell death (Fig 3). While expanding LYPAS capacity has been shown to suppress senescence or cell death, impaired LYPAS capacity might trigger either senescence or cell death based on the level of cell stress and the capacity of the lysosome. Based on this lysosomal-centric model of senescence/cell death (Fig 3), boosting the LYPAS capacity is a promising strategy for treating many age-related diseases, as well as LSDs that primarily affect a pediatric population (Fig 4).

The lysosomal model of senescence/cell death appears to be compatible with many studies. However, additional investigations are necessary to further test and improve this model (see Box 1). Many challenges remain, particularly considering the difficulty in identifying universal markers for senescence in various contexts. To strengthen the role of lysosomal overloading in senescence, it would be critical to distinguish cells in what we are calling early versus late senescence. Combining computational approaches and multiple senescence markers might help deconvolute the exact path leading to senescence and where lysosomal dysfunction impacts cell fate decisions.

Box 1. In need of answers

- i How are the different components of the lysosomal processing and adaptation system coordinated during a cellular stress response?
- ii How do diverse cellular stressors augment LYPAS capacity?
- iii How do overloaded and exhausted lysosomes signal outward and communicate with other components of the cell to initiate and maintain senescence?
- iv Is there an exact point of no return in senescence induction and what role do lysosomes play in this decision?
- v What role do lysosomes play in the maintenance and survival of senescent cells? Is lysosomal capacity a unique vulnerability of senescent cells that can be exploited therapeutically?
- vi Can manipulation of the LYPAS rejuvenate senescent cells or their clearance by the immune system?
- vii How does mild lysosomal stress benefit the cell, tissue, or organism?

Lysosomal processing and adaptation system-related studies have led to the discovery of many lysosomal-targeted therapeutics, several of which are being tested in clinical trials for age-related diseases, previously untreatable LSDs, or rare neurodegenerative conditions such as ALS or frontotemporal dementia. With more progress in the mechanistic understanding of LYPAS expansion capacity and LYPAS participation in aging and age-related diseases, additional therapeutic targets will likely be revealed. With the explosive development in drug design through deep learning, we expect to profoundly expand the list of small molecules that elevate LYPAS capacity, each potentially useful for treating a wide array of age-related diseases. Not bad for an organelle recovering from self-esteem issues!

Acknowledgements

We apologize to colleagues whose work could not be cited due to space limitations. This work was supported by Competitive Medical Research Fund (CMRF) of the University of Pittsburgh Medical Center (UPMC) Health System (JXT) and National Institutes of Health (NIH) grants 1K01AG075142 (JXT), 1R35GM150506 (JXT), 1R01HL142663 (TF), 1R01HL142589 (TF), U54AG075931 (TF), and P30AG024827 (TF).

Author contributions

Jay Xiaojun Tan: Conceptualization; writing – original draft; writing – review and editing. **Toren Finkel:** Writing – review and editing.

Disclosure and competing interests statement

JXT declares no competing interests. TF is a co-founder and stockholder in Generian Pharmaceuticals and Coloma Therapeutics.

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